1. **PmodKYPD.vhd**

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_ARITH.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;

entity PmodKYPD is
  Port (
    clk : in STD_LOGIC;
    JA : inout STD_LOGIC_VECTOR (7 downto 0);
    an : out STD_LOGIC_VECTOR (3 downto 0);
    seg : out STD_LOGIC_VECTOR (6 downto 0)
  );
end PmodKYPD;

architecture Behavioral of PmodKYPD is

  component Decoder is
    Port (
      clk : in STD_LOGIC;
      Row : in STD_LOGIC_VECTOR (3 downto 0);
      Col : out STD_LOGIC_VECTOR (3 downto 0);
      DecodeOut : out STD_LOGIC_VECTOR (3 downto 0)
    );
  end component;

  component DisplayController is
    Port (
      DispVal : in STD_LOGIC_VECTOR (3 downto 0);
      anode : out STD_LOGIC_VECTOR (3 downto 0);
      segOut : out STD_LOGIC_VECTOR (6 downto 0)
```

```
        );
    end component;

    signal Decode : STD_LOGIC_VECTOR (3 downto 0);

begin

    C0: Decoder
        port map (
            clk => clk,
            Row => JA(7 downto 4),
            Col => JA(3 downto 0),
            DecodeOut => Decode
        );

    C1: DisplayController
        port map (
            DispVal => Decode,
            anode => an,
            segOut => seg
        );

end Behavioral;
```

2. Decoder.vhd

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_ARITH.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;

entity Decoder is
    Port (
```

```
        clk : in STD_LOGIC;
        Row : in STD_LOGIC_VECTOR (3 downto 0);
        Col : out STD_LOGIC_VECTOR (3 downto 0);
        DecodeOut : out STD_LOGIC_VECTOR (3 downto 0)
    );
end Decoder;

architecture Behavioral of Decoder is

signal sclk : STD_LOGIC_VECTOR(19 downto 0);

begin

process(clk)
begin

if clk'event and clk = '1' then

-- 1ms
if sclk = "00011000011010100000" then

--C1
Col <= "0111";
sclk <= sclk + 1;

-- check row pins
elsif sclk = "00011000011010101000" then

--R1
if Row = "0111" then
DecodeOut <= "0001"; --1

--R2
```

```
elsif Row = "1011" then
DecodeOut <= "0100"; --4

--R3
elsif Row = "1101" then
DecodeOut <= "0111"; --7

--R4
elsif Row = "1110" then
DecodeOut <= "0000"; --0

end if;

sclk <= sclk + 1;

-- 2ms
elsif sclk = "00110000110101000000" then

--C2
Col <= "1011";
sclk <= sclk + 1;

-- check row pins
elsif sclk = "00110000110101001000" then

--R1
if Row = "0111" then
DecodeOut <= "0010"; --2

--R2
elsif Row = "1011" then
DecodeOut <= "0101"; --5
```

```
--R3
elsif Row = "1101" then
DecodeOut <= "1000"; --8

--R4
elsif Row = "1110" then
DecodeOut <= "1111"; --F

end if;

sclk <= sclk + 1;

--3ms
elsif sclk = "01001001001111100000" then

--C3
Col <= "1101";
sclk <= sclk + 1;

-- check row pins
elsif sclk = "01001001001111101000" then

--R1
if Row = "0111" then
DecodeOut <= "0011"; --3

--R2
elsif Row = "1011" then
DecodeOut <= "0110"; --6

--R3
elsif Row = "1101" then
DecodeOut <= "1001"; --9
```

```
--R4
elsif Row = "1110" then
DecodeOut <= "1110"; --E

end if;

sclk <= sclk + 1;

--4ms
elsif sclk = "01100001101010000000" then

--C4
Col <= "1110";
sclk <= sclk + 1;

-- check row pins
elsif sclk = "01100001101010001000" then

--R1
if Row = "0111" then
DecodeOut <= "1010"; --A

--R2
elsif Row = "1011" then
DecodeOut <= "1011"; --B

--R3
elsif Row = "1101" then
DecodeOut <= "1100"; --C

--R4
elsif Row = "1110" then
```

```
DecodeOut <= "1101"; --D

end if;

sclk <= "00000000000000000000";

else
sclk <= sclk + 1;

end if;

end if;

end process;

end Behavioral;
```

3. DisplayController.vhd

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_ARITH.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;

entity DisplayController is
  Port (
    DispVal : in STD_LOGIC_VECTOR (3 downto 0);
    anode : out STD_LOGIC_VECTOR (3 downto 0);
    segOut : out STD_LOGIC_VECTOR (6 downto 0)
  );
end DisplayController;

architecture Behavioral of DisplayController is
```

```
begin

anode <= "1110";

with DispVal select
segOut <= "1000000" when "0000", --0
        "1111001" when "0001", --1
        "0100100" when "0010", --2
        "0110000" when "0011", --3
        "0011001" when "0100", --4
        "0010010" when "0101", --5
        "0000010" when "0110", --6
        "1111000" when "0111", --7
        "0000000" when "1000", --8
        "0010000" when "1001", --9
        "0001000" when "1010", --A
        "0000011" when "1011", --B
        "1000110" when "1100", --C
        "0100001" when "1101", --D
        "0000110" when "1110", --E
        "0001110" when "1111", --F
        "0111111" when others;

end Behavioral;
```